

EEG-Based Error Related Potentials

Sprint Planning Document (Sprint 3)

Sprint Goal Backlog (Sprint 3)

Sprint Duration: February 25, 2026 – March 24, 2026

Team Members: Alexander Steffey, Ali Burkemper, Brody Greene, Sophia Kyemba, Will Mulhern

High-Level Project Overview

Project Mission:

To develop an accessible, low-cost framework for detecting, validating, and visualizing EEG-based Error-Related Potentials (ErrPs) using Python-based tools and consumer-grade EEG hardware.

Problems We Are Solving:

- Current pBCI (passive Brain-Computer Interface) development relies on expensive EEG equipment and proprietary frameworks
- Most existing ErrP validation systems require MATLAB-based tools
- There are limited frameworks for low-cost ErrP detection and visualization
- Researchers lack an accessible modular interface for EEG processing, detection, and visualization

Project Overview (High-Level Features):

1. ErrP Data Handling

- Load .set EEG files
- Handle continuous (non-epoched) EEG data
- Automatically label EEG data
- Allow file replacement and clearing

2. ErrP Visualization

- Display ErrP waveforms as time-series plots
- Display topographic maps (spatial brain activity)
- Display joint plots (time-series + topographic map)

- Update graphs dynamically
- Add event markers and response indicators

3. ErrP Detection

- Differentiate between error and non-error trials
- Implement real-time detection
- Implement real-time signal processing and classification

4. Accessibility & Interface

- GUI-based interface
- Modular code structure
- Standalone executable packaging

Sprint 3 Planning

Sprint 3 Goals:

1. Added ganglion device connectivity with proper flanker test
2. Complete device security and system testing
3. Provide all major documents overviewing project history
4. Complete general updates and bug fixes
5. Package final executable

1. Device connectivity and flanker task

- **Assigned:** Ali Burkemper
- Integrated a consumer-grade EEG device with a custom Flanker task to support real-time data collection
- Ensured the task followed standard cognitive experiment structure from stimulus presentation through feedback
- Enabled reliable data capture and automated formatting for later EEG analysis
- Provided a testing mode to allow demonstrations without requiring live hardware

2. Ganglion device security

- **Assigned:** Alexander Steffey
- Implemented safeguards to ensure secure and reliable connections with EEG hardware
- Improved system robustness through validation and controlled device access
- Ensured proper startup and shutdown behavior to protect recorded data

3. Code clean-up and restructuring

- **Assigned:** Brody Greene
- Improved code organization to support long-term maintainability
- Removed redundancies and clarified responsibilities across modules
- Standardized documentation and project structure

4. Project testing

- **Assigned:** Sophia Kyemba, Alexander Steffey
- Expanded testing coverage to verify system reliability
- Introduced automated and manual testing workflows
- Enabled safe testing of hardware-dependent features

6. Final bug and GUI fixes

- **Assigned:** Will Mulhern
- Resolved outstanding usability and stability issues
- Polished the graphical interface for consistency and responsiveness
- Improved support for multi-file workflows and exports

7. Documentation and presentation

- **Assigned:** Team-wide
- Completed final documentation covering system usage and setup
- Prepared presentation materials summarizing project outcomes
- Documented lessons learned for future development